

Cardiovascular Disease, Common Medications and Taking Blood Pressure

OMFS 512: Integrated Biomedical and Clinical Science

Rashidah Wiley, DDS, FAAOMP, D-ABOMP



Mission: To advance the science and art of health through education, service, scholarship and social accountability

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Learning Outcomes & Competency Statements

Learning Outcomes

1. Review of the common diseases of the cardiovascular system.
2. Examine the adverse reactions associated with antihypertensive drugs.
3. Discuss the dental consideration and modifications associated with cardiovascular disease.
4. Learn how to take blood pressure and what factors can influence blood pressure readings.

Competency statements met

1.3, 1.5, and 2.1

CARDIOVASCULAR DISEASE

THE WORLD'S NUMBER 1 KILLER

Cardiovascular diseases are a group of disorders of the heart and blood vessels, commonly referred to as **heart disease** and **stroke**.

20.5 deaths every year from CVD
MILLION

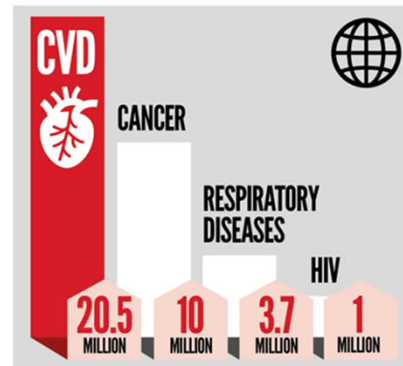


33% of all global deaths



>75% of CVD deaths take place in low- and middle-income countries

GLOBAL CAUSES OF DEATH



RISK FACTORS FOR CVD



Sources: World Health Organization;
IHME, Global Burden of Disease

info@worldheart.org
www.worldheart.org

f worldheartfederation
worldheartfed
@ worldheartfederation

world-heart-federation.org/resource/cardiovascular-disease-infographic/

ledicine

Categories of Heart Disease

- Hypertension
- Ischemic Heart Disease (IHD)/Coronary Artery Disease (CAD)
- Valvular Heart Disease
- Dysrhythmias/Arrhythmias
- Heart Failure
- Infective Endocarditis*
- Congenital Heart Disease*

Hypertension

- Abnormal elevation of arterial blood pressure
- Classifications
 - Essential - "Primary" "Idiopathic"
 - Initially asymptomatic
 - Symptoms – Nosebleeds, dizziness, and headaches
 - Secondary
 - Indefinable cause – renal failure, drug abuse, etc.
 - Malignant
 - Sudden rise in blood pressure

Hypertension

Blood Pressure Categories



BLOOD PRESSURE CATEGORY	SYSTOLIC mm Hg (upper number)		DIASTOLIC mm Hg (lower number)
NORMAL	LESS THAN 120	and	LESS THAN 80
ELEVATED	120 – 129	and	LESS THAN 80
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 1	130 – 139	or	80 – 89
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 2	140 OR HIGHER	or	90 OR HIGHER
HYPERTENSIVE CRISIS (consult your doctor immediately)	HIGHER THAN 180	and/or	HIGHER THAN 120

Risk Factors

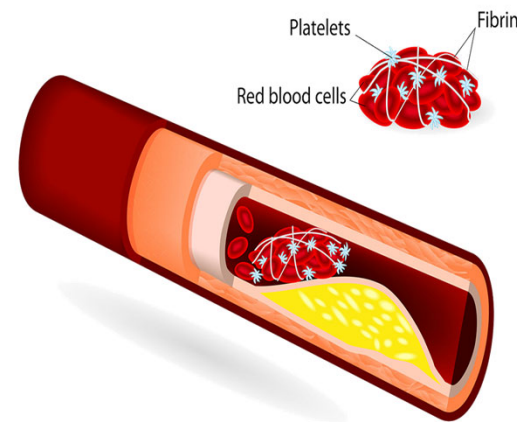
Table 2.1. Risk Factors for Cardiovascular Disease

Modifiable	Nonmodifiable
High blood pressure	Age
Atherosclerosis/dyslipidemia	Sex
Diabetes	Family history
Tobacco smoking	
Obesity/diet	
Inactivity	
Stress	
Alcohol use	

Ischemic Heart Disease

" Coronary artery disease"

- Atheroma forms within the vascular wall of arteries leading to hardening of blood vessels
(atherosclerosis)
- Clot formation
- Embolism



www.hri.org.au/health/learn/cardiovascular-disease/blood-clots

Ischemic Heart Disease

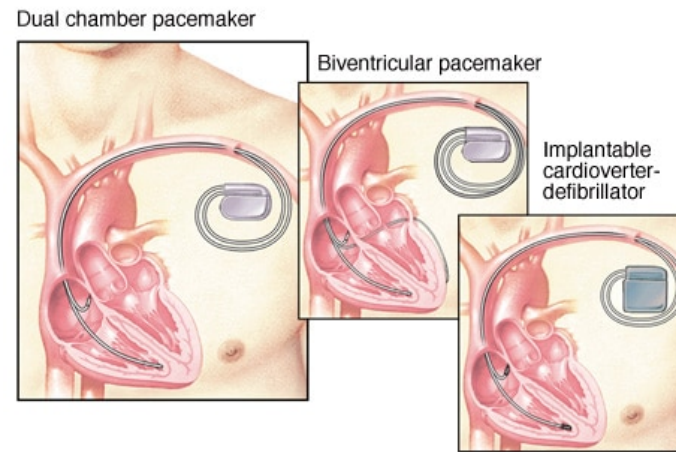
- Coronary artery blockage
 - **Stable angina pectoris**
 - Chest pain and shortness of breath lasting 5-15 minutes upon exertion.
 - **Unstable angina pectoris**
 - Chest pain and shortness of breath which is unpredictable and harder to control.
 - **Myocardial infarction (MI)**
 - Symptoms vary in affected patients

Arrhythmias (Dysrhythmias)

- Disruption of the electrical impulses or conduction in the heart
- Increased risk of clot formations and thromboembolic events
 - **Atrial fibrillation**
 - Most common type of arrhythmia
 - Pt. May take anticoagulant therapy - Coumadin
 - Tachycardia
 - Supraventricular tachycardia
 - Ventricular fibrillation
 - Most common risk of sudden death

Atrial Fib

- Surgically implanted devices which help normalize the electric impulses of the heart.
 - Pacemakers
 - Implantable Cardioverter-Defibrillators
- Electronic dental devices may interfere with these devices.



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Dental Instruments to Avoid in Patients with Arrhythmias



Electrosurgery Machine



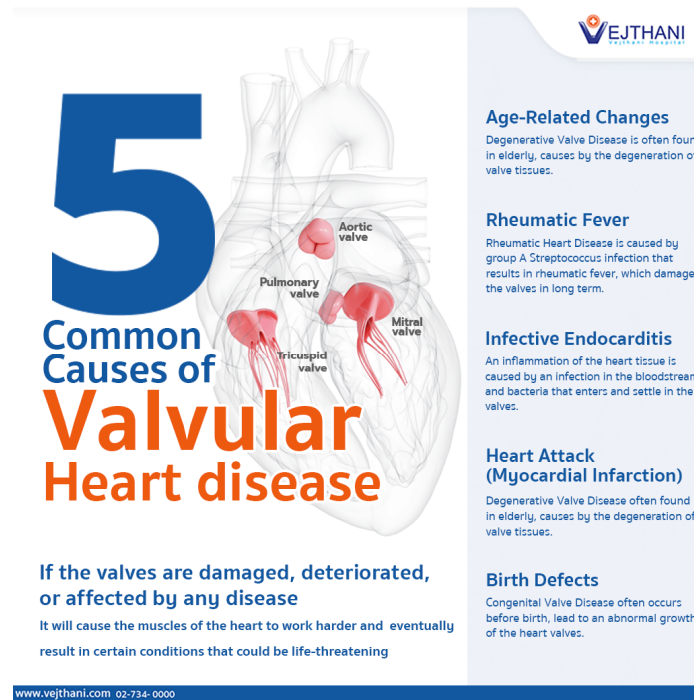
Cavitron



Dental Laser

Valvular Heart Disease

- Stenosis or insufficiency of the heart valves associated:
- **Rheumatic fever**
- **Infective endocarditis**
- Age
- Myocardial Infarctions
- Birth defects



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Common Causes of
Valvular Heart disease

Age-Related Changes
Degenerative Valve Disease is often found in elderly, caused by the degeneration of valve tissues.

Rheumatic Fever
Rheumatic Heart Disease is caused by group A Streptococcus infection that results in rheumatic fever, which damages the valves in long term.

Infective Endocarditis
An inflammation of the heart tissue is caused by an infection in the bloodstream and bacteria that enters and settle in the valves.

Heart Attack (Myocardial Infarction)
Degenerative Valve Disease often found in elderly, caused by the degeneration of valve tissues.

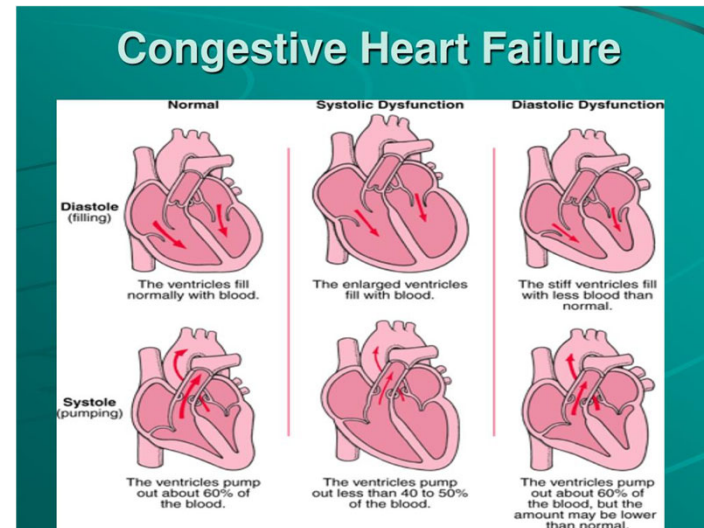
Birth Defects
Congenital Valve Disease often occurs before birth, lead to an abnormal growth of the heart valves.

If the valves are damaged, deteriorated, or affected by any disease
It will cause the muscles of the heart to work harder and eventually result in certain conditions that could be life-threatening

www.vejthani.com 02-734-0000

Heart Failure

- Occurs when the heart cannot maintain adequate circulation
 - Pump failure
 - Increased workload
- Clinical symptoms
 - Dyspnea
 - Orthopnea
 - Fatigue
 - Unable to withstand long periods of physical activity
 - Peripheral and pitting edema



Clinical Symptoms of Heart Failure

RIGHT VS. LEFT-SIDED HEART FAILURE
Cor Pulmonale

RIGHT-SIDED HEART FAILURE	LEFT-SIDED HEART FAILURE
S SWELLING OF THE LEGS & HANDS 	D DYSPNEA (SHORTNESS OF BREATH) 
W WEIGHT GAIN	R RALES (CRACKLES)
E EDEMA (PITTING) 	O ORTHOPNEA
L LARGE NECK VEINS (JUGULAR VEIN DISTENTION) 	W WEAKNESS/FATIGUE 
L LETHARGY OR FATIGUE	N NOCTURNAL PAROXYSMAL DYSPNEA
I IRREGULAR HEART RATE 	I INCREASED HEART RATE
N NOCTURIA 	N NAGGING COUGH (FORTHY, BLOOD TINGED SPUTUM) 
G GIRTH (ASCITES) 	G GAIN WEIGHT

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Antihypertensive Medications

- Systemic manifestations
 - Excessive or spontaneous bleeding
 - Hypersensitivity reactions
- Oral reactions
 - Gingival hyperplasia
 - Ulcers
 - Lichenoid drug reaction



Table 2.2. Common Medications for Cardiovascular Disease and Oral Adverse Drug Reaction/Dental Implication

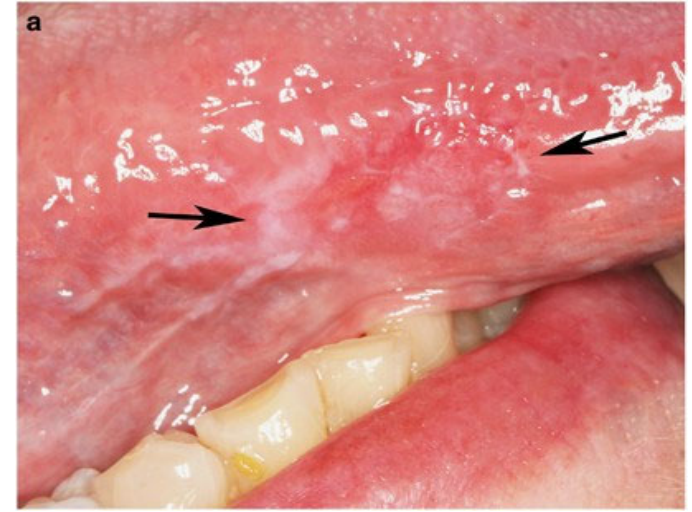
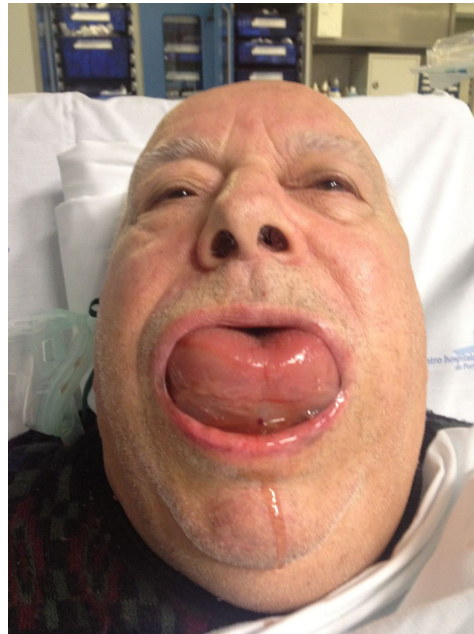
Cardiovascular Drug Category	Culprit Drug	Oral Adverse Drug Reaction/ Dental Implication and Likelihood of Association
Alpha-adrenergic blockers	Class effect Methyldopa	Est: dry mouth Pos: lichenoid drug eruption
Beta-adrenergic blockers	Class effect/unspecified Labetolol/unspecified Atenolol, oxprenolol, practolol, propranolol Propranolol Practolol Carvediol Propranolol (sublingual)	Est: dry mouth, angioedema Prob: aphthae/ulcers Pos: lichenoid drug eruption Pos: thrombocytopenia Pos: oculo-mucocutaneous syndrome Pos: SJS Pos: mouth paresthesia
Angiotensin-converting enzyme (ACE) inhibitors	Class effect Class effect Captopril, enalapril Captopril Captopril, enalapril, lisinopril Lisinopril Captopril	Est: angioedema Est: neutropenia/agranulocytosis Est: taste disturbances Prob: aphthae/ulcers; pemphigus Prob: scalded mouth syndrome Prob: dry mouth Pos: lichenoid drug eruption
Angiotensin II receptor blocker	Losartan	Prob: angioedema

Table 2.2. (Continued)

Cardiovascular Drug Category	Culprit Drug	Oral Adverse Drug Reaction/ Dental Implication and Likelihood of Association
Anti-arrhythmics, Class I (sodium channel blockers)	Class effect	<i>Est:</i> dry mouth
	Phenytoin	<i>Est:</i> gingival overgrowth; hypersensitivity reaction syndrome
	Phenytoin	<i>Prob:</i> agranulocytosis; SJS, TEN
	Quinidine	<i>Prob:</i> thrombocytopenia
	Quinidine	<i>Pos:</i> FDE
Anti-arrhythmics, Class III (potassium channel blockers)	Amiodarone	<i>Prob:</i> angioedema
	Amiodarone	<i>Pos:</i> taste disturbance
Calcium channel blockers	Class effect	<i>Est:</i> gingival overgrowth, dry mouth, taste disturbances
	Diltiazem, verapamil	<i>Pos:</i> aphthae/ulcers, EM, SJS, TEN
	Nifedipine, diltiazem	<i>Pos:</i> angioedema
	Amlodipine	<i>Pos:</i> lichenoid drug eruption
Diuretics	Class effect	<i>Est:</i> dry mouth
	Amiloride, furosemide, hydrochlorothiazide	<i>Prob:</i> agranulocytosis, thrombocytopenia
	Amiloride, spironolactone	<i>Prob:</i> taste disturbance
	Bendroflumazide, furosemide/ spironolactone	<i>Uncertain/Pos:</i> lichenoid drug eruption
	Hydrochlorothiazide, furosemide	<i>Pos:</i> EM, SJS, TEN, drug hypersensitivity reaction
	Unspecified	<i>Pos:</i> angioedema
Potassium channel opener	Nicorandil	<i>Prob:</i> aphthae/ulcers
Direct-acting peripheral vasodilator	Hydralazine	<i>Prob:</i> lupus erythematosus
Lipid regulators (statins)	Simvastatin	<i>Pos:</i> cheilitis, lichenoid drug eruption
Platelet inhibitors	Aspirin	<i>Prob:</i> FDE
	Aspirin	<i>Pos:</i> angioedema
Anticoagulants	Warfarin, dabigatran, rivaroxaban	<i>Est:</i> impaired hemostasis

Est, established drug reaction; *Prob,* probable drug reaction; *Pos,* possible drug reaction; SJS, Stevens-Johnson syndrome; TEN, toxic epidermal necrolysis; FDE, fixed drug reaction; EM, erythema multiforme.
Modified from Torpet et al.¹⁰

Oral Manifestations of Drug Reactions



Dental Consideration

- Review the medical history
 - Identification of patients with hypertension
 - Be familiar the patients' medications
 - Take blood pressure
- Anxiety and pain control
 - Risk associated with surgical procedures
 - Local anesthesia
- Chair positioning

Dental Modifications

- Blood Pressure in CNU CDM Clinic
 - 140/90
 - Re-take blood pressure
 - Most dental treatment can still take place with modification
 - **160/95**
 - Re-take blood pressure
 - If constant, this is the **CDM CUTOFF** – no treatment
 - **180/100**
 - Asymptomatic – No treatment, refer to PCP or Urgent care
 - Symptomatic – No treatment and send to ER

Dental Modifications

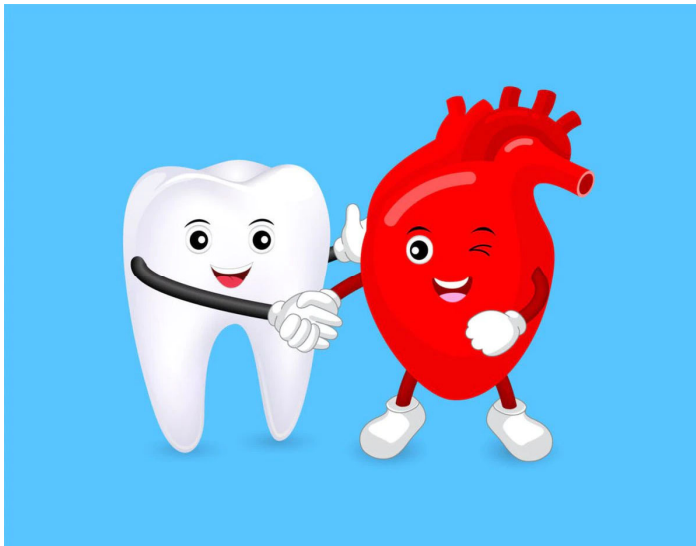
- Delaying non-urgent invasive dental treatment after myocardial infarction (3-6 months) .
- Antibiotic prophylaxis

TABLE 1. Indications for Antimicrobial Prophylaxis¹

Prosthetic cardiac valves/material	• Transcatheter implanted prosthetic valves • Annuloplasty, rings, or clips • Left ventricular assist devices or implantable heart
Congenital heart disease (CHD)	• Unrepaired cyanotic congenital, including palliative shunts and conduits • Completely repaired defect with prosthetic material or device within 6 months after the procedure • Repaired CHD with residual defects at or adjacent to the site of prosthetic material • Surgical or transcatheter pulmonary artery valve or conduit placement
Previous, relapse, or recurrent infective endocarditis	
Cardiac transplant recipients who develop cardiac valvulopathy	

The Importance of Blood Pressure in Dental Care: How to Correctly Take Blood Pressure

Hypertension and Dentistry



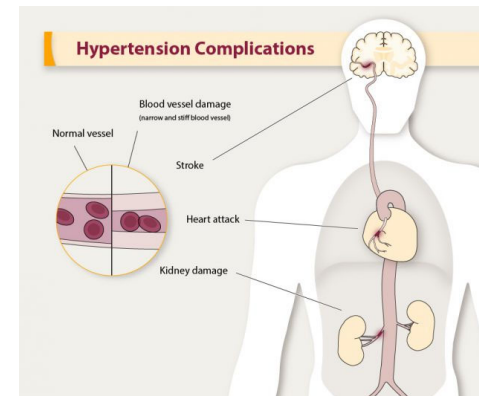
www.sun-sentinel.com/florida-jewish-journal/fl-jj-oral-health-heart-disease-not-well-understood

- Hypertension is one of the most common systemic diseases encountered in dentistry.
- Hypertension has a great impact on dental health
 - Missed appointments
 - Treatment modifications
 - Deferment of dental treatments

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Blood pressure readings and dental practice

- What does my patient's blood pressure readings mean?
 - Blood pressure of 140/90
 - Stage II hypertension
 - **Blood pressure of 160/100**
 - Greater risk of adverse events
 - Blood pressure of 180/120
 - Hypertensive crisis
 - Hypertensive urgency "Uncontrolled hypertension"
 - Hypertensive emergency



Way to Acquire Blood Pressure



How to take blood pressure

BP Measurement Checklist

Measure blood pressure of all adults ≥ 18 years.

No talking during and between measurements

Back supported

Cuff at heart level

Use correct cuff size and positioning.

Arm supported

Small Medium Large

1. Acceptable cuff bladder length is at least 80% of the patient's arm circumference.

2. Acceptable cuff bladder width is at least 40% of the patient's upper arm circumference.

3. Ensure cuff is on bare arm or over thin layer of clothing. Avoid bunching of clothes under cuff.

4. Empty bladder before each reading.

Record exact reading from digital device. Don't round.

Legs uncrossed and feet supported

Avoid exercise, tea/coffee, smoking in the last 30 minutes.
Patient should rest comfortably and quietly for 5 minutes before the reading.

How to take blood pressure

1. Place BP cuff 1 ½ inches above antecubital space
2. Center over brachial artery
3. Check pulse – if irregular use manual BP cuff only
4. Keep thumb off the bell of stethoscope
5. Inflate to 20- 30 mmHg above where pulse is no longer palpable
6. Slowly deflate at a steady rate, 2-3 mmHg per heartbeat
7. Listen for first of two or more beats (Korotkoff sounds) SYSTOLIC BP
8. Last muffled beat (Korotkoff sound) DIASTOLIC BP
9. If BP is elevated, repeat after 1-2 minutes
10. Report to PCP for follow up if elevated BP

<https://youtu.be/mTTX5AhPT2E>

www.health.nd.gov/sites/www/files/documents/Files/HSC/HP/OralHealth

External Factors Affecting Blood Pressure

- Physical activity
- The use of stimulants
- Mental and emotion stress
- Fear and anxiety
 - White-coat syndrome



Clinical Judgement



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Resources

